RP2350A and core supply

U1
RP2350A
QSPI_SS
QSPI_SD3
QSPI_SD2
QSPI_SD1
QSPI_SD0
QSPI_SCLK
VREG_LX
VREG_VIN
VREG_FB
ADC_AVDD
DVDD
IOVDD
USB_OTP_VDD
QFN-60
RUN
USB_DM
USB_DP
FLASH_CS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
XIN
XOUT
SWCLK
SWDIO
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
PSRAM_CS
SPI0_SCK
I2C_SDA
I2C_SCL
SD_MOSI
SD_MISO
SD_CLK
CHARGE_DISABLE
PSRAM_SCK
SD_DETECT
RGB_DATA
SPI0_CSn
SPI0_TX
SPI0_RX
RADIO_CTRL
UART_TX
UART_RX
UART_RTS
UART_CTS
SPI0_CSn
SPI0_TX
SPI0_RX
SPI0_SCK
GAUGE_SDA
GAUGE_SCL
SPI0_SCK
SPI0_TX
SPI0_RX
SPI0_CSn
ADC3
ADC2
ADC1
ADC0

C1
100 nF
C2
100 nF
R1
10 kΩ
FLASH_CS
R2
1.8 kΩ
PSRAM_CS

SW1
BOOT
SW2
RESET
BOOT_SWITCH
RUN

Y1
12 MHz
XIN
XTAL_OUT

R4
10 kΩ
R5
1 kΩ
C3
15 pF
C4
15 pF
L1
3.3 µH
R6
33 Ω
C5
4.7 µF
C6
22 µF
C7
4.7 µF
C8
4.7 µF
C9
100 nF
C10
100 nF
C11
100 nF
C12
100 nF
C13
100 nF
C14
100 nF
C15
100 nF
C16
100 nF
C17
100 nF
C18
100 nF
C19
100 nF
C20
100 nF

16 MB flash + 8 MB PSRAM

U2
16 MB flash
U3
8 MB PSRAM

Firmware: GPIO0 = QMI CS1n. PSRAM must be initialised explicitly.
RP2350A production stepping A4 preferred. All GPIOs are 3.3 V only.

PWR_FLAG
GND
VBUS
CORE_1V1
VREG_AVDD
BUCK_VINA

USB/battery power and charging

U5
Power-path charger
U6
3.3 V buck-boost
U7
Wi-Fi + BLE
U8
Battery voltage / SOC
CHARGE_DISABLE
CHG_TMR
USB_500_EN
CHG_ILIM
CHG_ISET
BAT_TS
USB_PGOOD_N
GAUGE_SCL
GAUGE_SDA
BUCK_L1
BUCK_L2
BUCK_VINA
VBUS
1V1
BAT
L2
2.2 µH
C21
1 µF
C22
10 µF
C23
10 µF
C24
10 µF
C25
100 nF
R36
100 Ω
C26
22 µF
C27
22 µF
R11
8.66 kΩ
R12
3.48 kΩ
R13
10 kΩ
R14
68.1 kΩ
R16
10 kΩ
R17
10 kΩ
R18
10 kΩ
R19
10 kΩ
R20
3.3 kΩ
R21
3.3 kΩ
C28
100 nF
J2
LiPo

Wi-Fi and Bluetooth LE companion

RADIO_EN
RADIO_CTRL
RADIO_RX
RADIO_TX
RADIO_BOOT
RADIO_IO0
RADIO_IO2
R22
100 kΩ
R35
1 kΩ
R24
10 kΩ
C30
10 µF
C32
22 µF
C29
1 µF
R23
10 kΩ
R25
10 kΩ
C31
100 nF
SW3
RADIO BOOT
J3
Radio UART

USB default 100 mA; charging disabled, radio in reset.
Firmware selects 500 mA and enables charging only after enumeration.
Protected 1S LiPo only. TS uses fixed 10 kOhm: no pack temperature sensing.

UART0 GPIO12–15 to ESP-AT command port GPIO6/7/4/5.
GPIO16 controls radio EN. UART0 on J3 is for radio programming.
Antenna overhangs the north board edge. Keep enclosure metal away.

USB-C device port

J1
USB-C
USB ESD
CC1
CC2
USB_D_P
USB_D_N
R7
5.1 kΩ
R8
5.1 kΩ
R9
27 Ω
R10
27 Ω

GPIO / SWD / test access

J6
GPIO
J7
SWD

Storage and sensor interfaces

J4
microSD
J5
Qwiic / STEMMA QT
D1
RGB V5
R26
10 kΩ
R27
10 kΩ
R28
10 kΩ
R29
10 kΩ
R30
10 kΩ
R31
10 kΩ
C33
10 µF
C34
100 nF
R32
5.1 kΩ
R33
5.1 kΩ
R34
330 Ω
C35
100 nF

Qwiic: pin 1 GND, 2 3V3, 3 SDA, 4 SCL.
Battery: pin 1 GND, pin 2 BAT. Verify cable polarity.
External header supplies are references, never power inputs.

TP1
VBUS
TP2
SYS
TP3
+3V3
TP4
GND
TP5
BAT
TP6
CHG_N
TP7
USB_PGOOD_N
TP8
CORE_1V1

H1
M2.5
H2
M2.5
H3
M2.5
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial

Prototype development board
Sheet: /
File: sprocket2350.kicad_sch

Title: Sprocket 2350 | USB-C, LiPo, Wi-Fi/BLE, PSRAM

Size: A3	Date:
KiCad E.D.A. 10.99.0-6544~g8c8f8ac1452	Rev: A
	Id: 1/1